

1. Define Artificial Intelligence and explain its types.

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence such as learning, reasoning, and decision-making. Types include Narrow AI (single-task systems), General AI (human-level intelligence), and Super AI (beyond human intelligence, theoretical).

Real-World Example: Virtual assistants like Alexa are examples of Narrow AI.

2. Explain the characteristics of AI.

AI systems can learn from data, reason logically, solve problems, adapt to new situations, and make decisions. They mimic human intelligence using algorithms and data processing.

Real-World Example: Recommendation systems on Netflix learn user preferences and suggest movies.

3. What is an Intelligent Agent?

An Intelligent Agent is an entity that perceives its environment through sensors and acts upon that environment using actuators to achieve goals efficiently.

Real-World Example: A robotic vacuum senses dirt and moves to clean it.

4. Explain PEAS with example.

PEAS stands for Performance measure, Environment, Actuators, and Sensors. It is used to describe an agent's task environment.

Real-World Example: For a self-driving car: Performance (safe driving), Environment (roads), Sensors (cameras), Actuators (steering, brakes).

5. Describe types of environments in AI.

Environments can be fully/partially observable, deterministic/stochastic, static/dynamic, and discrete/continuous. These factors affect agent design.

Real-World Example: Chess is fully observable and deterministic.

6. Explain Problem-Solving Agent.

A Problem-Solving Agent formulates a goal, searches possible solutions, and selects the best action to achieve the goal.

Real-World Example: Google Maps finding shortest route.

7. Explain Breadth First Search (BFS).

BFS explores nodes level by level. It guarantees the shortest path but consumes more memory.

Real-World Example: Used in finding shortest path in maps.

8. Explain Depth First Search (DFS).

DFS explores deeply along one branch before backtracking. It uses less memory but may not find optimal solution.

Real-World Example: Maze-solving algorithms often use DFS.

9. What is Heuristic Function?

A heuristic function estimates the cost from current state to goal state to speed up search.

Real-World Example: Straight-line distance in GPS navigation.

10. Compare BFS and DFS.

BFS finds shortest path but uses more memory. DFS uses less memory but may not find shortest path. BFS is complete; DFS may not be complete in infinite trees.

Real-World Example: Route planning prefers BFS-like strategy.

11. Explain Perception-Action Cycle.

An agent continuously senses environment, processes information, acts, and then senses again in a loop.

Real-World Example: Thermostat adjusting temperature.

12. What is Planning in AI?

Planning is generating a sequence of actions to achieve a goal from initial state.

Real-World Example: Robot planning steps to assemble a product.

13. Explain STRIPS.

STRIPS is a planning system where actions are defined by preconditions and effects to reach goal states.

Real-World Example: Robot checks if door is open before entering room.

14. Explain Goal Stack Planning.

It works backward from goal by breaking it into smaller sub-goals and solving them sequentially.

Real-World Example: To travel abroad, first get passport, then visa.

15. What is Nonlinear Planning?

Nonlinear planning allows actions to be arranged flexibly instead of strict sequence.

Real-World Example: Project management tools that allow parallel tasks.

16. Explain Robot Architecture.

Robot architecture includes sensors, control system (processor), and actuators for movement and actions.

Real-World Example: Industrial robots assembling cars.

17. Define Natural Language Processing (NLP).

NLP enables machines to understand and process human language.

Real-World Example: Chatbots handling customer support.

18. Explain Natural Language Understanding (NLU).

NLU focuses on understanding meaning and context of language.

Real-World Example: Voice assistants understanding spoken commands.

19. Explain Syntax and Semantics.

Syntax deals with grammar rules; semantics deals with meaning of words and sentences.

Real-World Example: Correct grammar ensures correct interpretation.

20. What is Disambiguation?

Disambiguation resolves multiple meanings of a word based on context.

Real-World Example: Bank could mean river bank or financial bank.

21. Explain Inference Engine.

Inference engine applies logical rules to knowledge base to derive conclusions in expert systems.

Real-World Example: Medical diagnosis systems.

22. Limitations of Syntax-Only Analysis.

Grammar alone cannot capture context or real meaning; semantic analysis is needed.

Real-World Example: 'He saw the man with telescope' has multiple meanings.

23. Role of Semantic Knowledge.

Semantic knowledge helps systems understand meaning, context, and relationships between words.

Real-World Example: Search engines understanding user intent.

24. Define Crisp Set.

Crisp set has clear membership: either true (1) or false (0).

Real-World Example: Age ≥ 18 is adult.

25. Define Fuzzy Set.

Fuzzy set allows partial membership between 0 and 1 to handle uncertainty.

Real-World Example: 'Tall person' can vary by degree.

26. Explain Fuzzification and Defuzzification.

Fuzzification converts crisp input into fuzzy values. Defuzzification converts fuzzy output into crisp value.

Real-World Example: AC adjusts cooling based on fuzzy temperature input.

27. What is Fuzzy Logic?

Fuzzy logic is reasoning method based on degrees of truth rather than binary logic.

Real-World Example: Washing machines adjust washing time automatically.

28. Explain Neural Networks.

Neural networks are computational models inspired by human brain neurons, consisting of layers and weighted connections.

Real-World Example: Image recognition systems.

29. Explain Perceptron Learning Rule.

Perceptron adjusts weights based on error between predicted and actual output to improve accuracy.

Real-World Example: Spam detection improving after mistakes.

30. Explain Backpropagation Algorithm.

Backpropagation is training method where error is propagated backward through network to update weights using gradient descent.

Real-World Example: Face recognition improving accuracy after training.

SHORT REVISION NOTES

- AI enables intelligent decision-making.
- Agents use PEAS model.
- BFS = shortest path; DFS = memory efficient.
- Heuristics speed up search.
- Planning includes STRIPS and Goal Stack.
- NLP helps understand language.
- Fuzzy logic handles uncertainty.
- Neural networks learn using backpropagation.